DO NOW — RECALL

Lesson 096 · Little Man Computer: Instructions and the A

Name: _____ Class: _____ Date: _____

1. What is the ACC for?

2. What does STA X do?

3. What does INP do?

4. Half adder A1,B1: what Sum and Carry?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Stores current/intermediate calculation.

2. Model answer

Stores current ACC into mailbox X.

3. Model answer

Reads input into ACC, replacing its former value.

4. Model answer

Sum0 (XOR); Carry1 (AND).

By the end, you can…

• LMC is a teaching model of a stored-program accumulator processor: mailboxes hold instructions or data, and ACC holds the immediate working value.

• INP replaces ACC; STA copies ACC into a mailbox; LDA copies a mailbox's contents into ACC; ADD adds mailbox contents to ACC.

• OUT writes current ACC to output. Trace ACC separately from mailbox contents and PC instruction addresses.

Vocabulary — know these terms

LMC — Little Man Computer, simplified stored-program model.

mailbox — Memory location.

ACC — Accumulator for working results.

STA — Store ACC into memory.

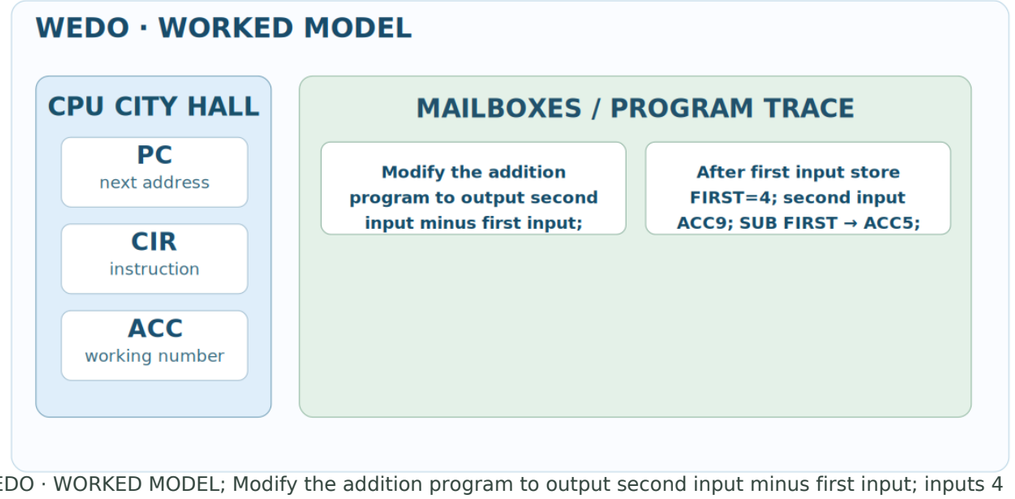
LDA — Load ACC from memory.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Modify the addition program to output second input minus first input; inputs 4 then 9.

After first input store FIRST=4; second input ACC9; SUB FIRST → ACC5; OUT5.



WEDO · WORKED MODEL; Modify the addition program to output second input minus first input; inputs 4 then 9.; After first input store FIRST=4; second input ACC9; SUB FIRST → ACC5; OUT5.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Follow the reasoning
After first input store FIRST=4; second input ACC9; SUB FIRST → ACC5; OUT5.

NEW LEARNING

Read, annotate and explain

Observe

LMC is a teaching model of a stored-program accumulator processor: mailboxes hold instructions or data, and ACC holds the immediate working value.

Apply

INP replaces ACC; STA copies ACC into a mailbox; LDA copies a mailbox's contents into ACC; ADD adds mailbox contents to ACC.

Evaluate

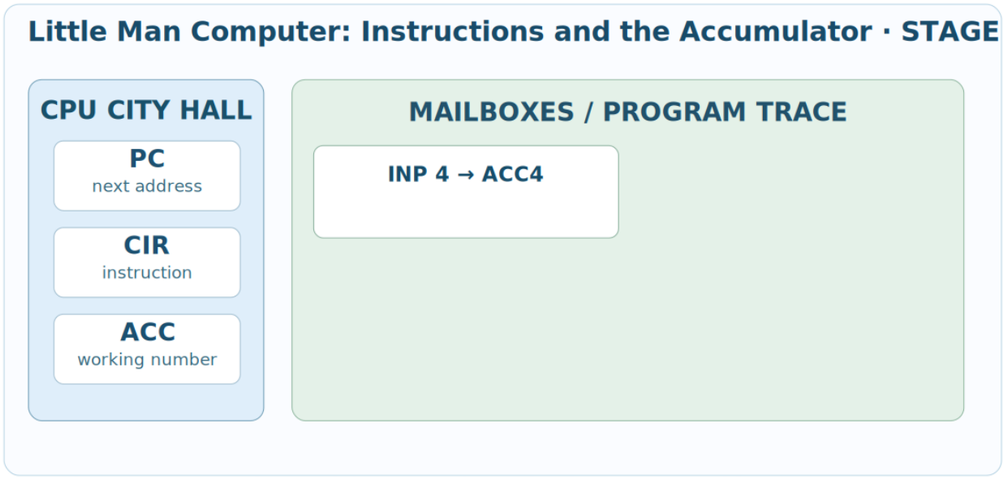
OUT writes current ACC to output. Trace ACC separately from mailbox contents and PC instruction addresses.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

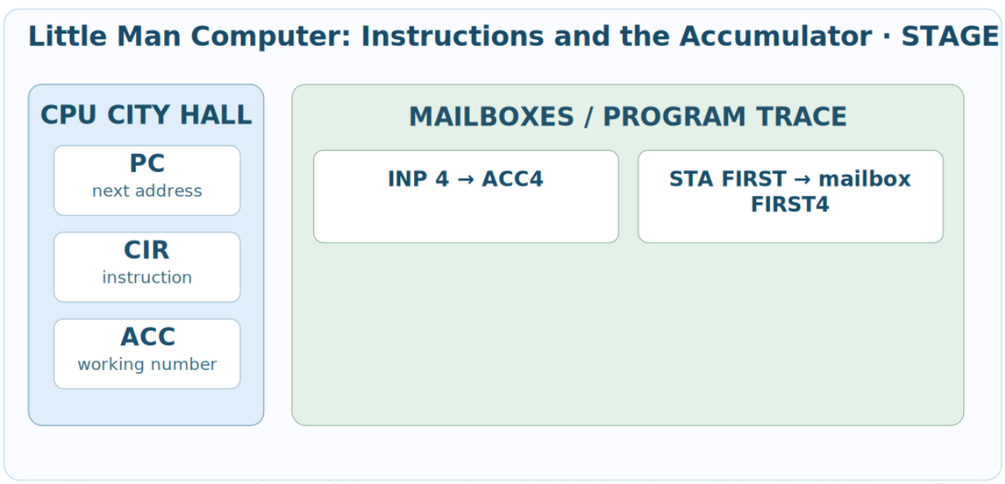
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Little Man Computer: Instructions and the Accumulator · STAGE 1; INP 4 → ACC4

New Learning



Little Man Computer: Instructions and the Accumulator · STAGE 2; INP 4 → ACC4; STA FIRST → mailbox FIRST4

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

STA FIRST overwrites ACC with the mailbox's value.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

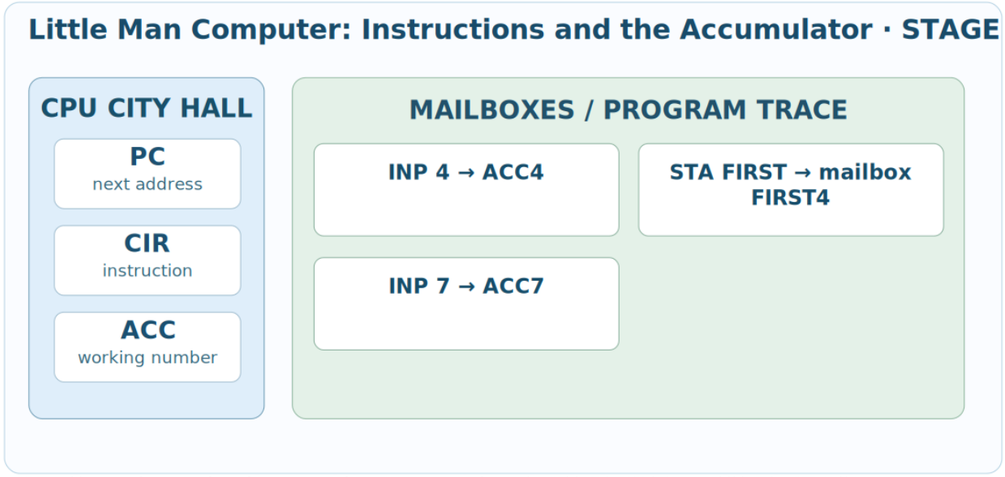
Question 3

Compare LDA FIRST, STA FIRST and ADD FIRST in relation to ACC and mailbox; give distinct effects. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

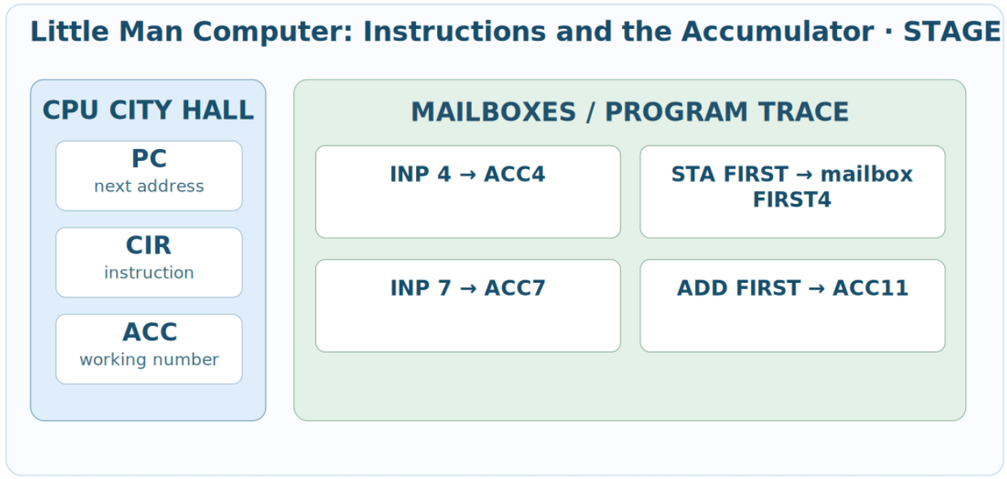


Little Man Computer: Instructions and the Accumulator · STAGE 3; INP 4 → ACC4; STA FIRST → mailbox FIRST4; INP 7 → ACC7

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Little Man Computer: Instructions and the Accumulator · STAGE 4; INP 4 → ACC4; STA FIRST → mailbox FIRST4; INP 7 → ACC7; ADD FIRST → ACC11

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace program above for inputs 4,7: ACC after each INP, STA and ADD; output. [6]

Question 2

Write an LMC program that reads one integer, stores it as N, then loads and outputs N. [6]

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

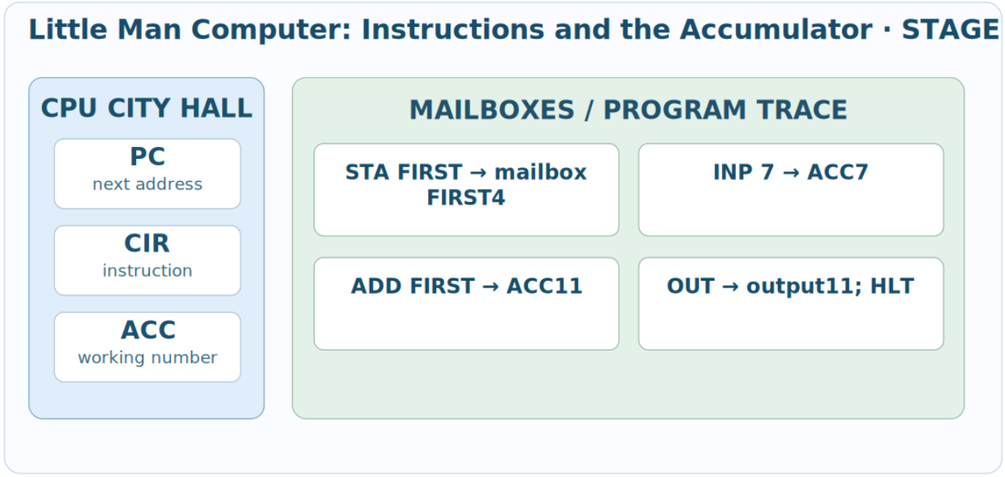
Guided 1 — TRY

Modify the addition program to output second input minus first input; inputs 4 then 9.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Little Man Computer: Instructions and the Accumulator · STAGE 5; INP 4 → ACC4; STA FIRST → mailbox FIRST4; INP 7 → ACC7; ADD FIRST → ACC11; OUT → output11; HLT

I DO — WORKED EXAMPLE

Follow the worked method

Inputs 6 then 3 to the shown program. What is FIRST, ACC after ADD, and OUT?

Follow the reasoning

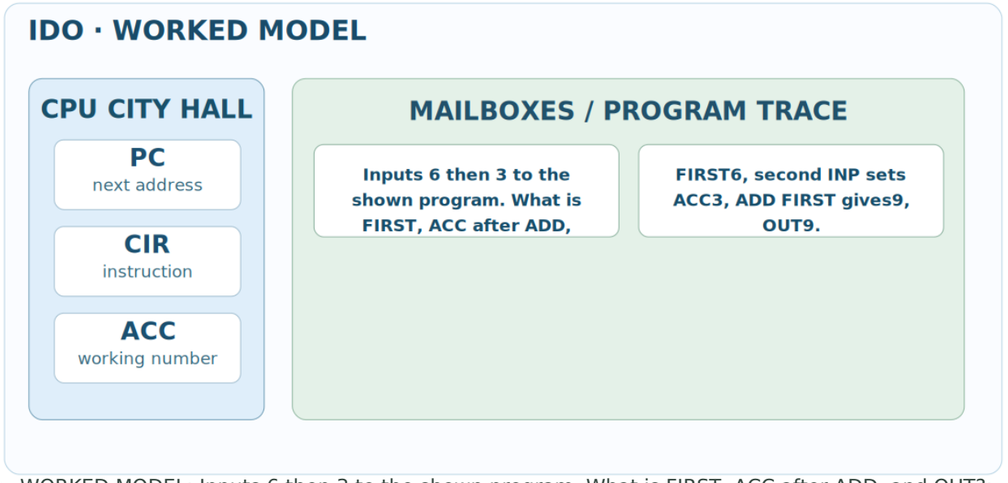
FIRST6, second INP sets ACC3, ADD FIRST gives9, OUT9.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Inputs 6 then 3 to the shown program. What is FIRST, ACC after ADD, and OUT?

FIRST6, second INP sets ACC3, ADD FIRST gives9, OUT9.



IDO · WORKED MODEL; Inputs 6 then 3 to the shown program. What is FIRST, ACC after ADD, and OUT?; FIRST6, second INP sets ACC3, ADD FIRST gives9, OUT9.